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Vehicle sensor assembly

Abstract

A sensor assembly includes a bracket fixed to a frame of a vehicle, two environmental sensors fixed to the bracket, and an insulating plate positioned between the environmental sensors and an inside of the vehicle. The sensor assembly may enhance the operation of the environmental sensors. Mounting both environmental sensors to the same bracket can permit more stringent position tolerances to be met by the environmental sensors compared to independently mounting the environmental sensors to the frame of the vehicle. Moreover, the bracket can provide a reference for locating the insulating plate. Additionally, the insulating plate can isolate the environmental sensors from heat generated by the vehicle, permitting the environmental sensors to more easily be kept within a temperature range for reliable operation. Finally, the same insulating plate can be used for both environmental sensors, in part due to the common position from the bracket.

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Background/Summary

BACKGROUND

(1) Vehicles often include a variety of sensors. Some sensors detect internal states of the vehicle, for example, wheel speed, wheel orientation, and engine and transmission variables. Some sensors detect the position or orientation of the vehicle, for example, global positioning system (GPS) sensors; accelerometers such as piezo-electric or microelectromechanical systems (MEMS); gyroscopes such as rate, ring laser, or fiber-optic gyroscopes; inertial measurements units (IMU); and magnetometers. Some sensors detect the external world, for example, radar sensors, scanning laser range finders, light detection and ranging (lidar) devices, and image processing sensors such as cameras. Radar sensors transmit radio waves and receives reflections of those radio waves to detect physical objects in the environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a portion of an example vehicle.
- (2) FIG. 2 is a rear perspective view of a sensor assembly in the vehicle.
- (3) FIG. 3 is a perspective view of the sensor assembly in the vehicle.

DETAILED DESCRIPTION

- (4) A sensor assembly includes a bracket fixed to a frame of a vehicle, two environmental sensors fixed to the bracket, and an insulating plate positioned between the environmental sensors and an inside of the vehicle.
- (5) The sensor assembly may further include a blower fixed to the bracket and positioned to draw air across the environmental sensors. The blower may be positioned on an opposite side of the insulating plate as the environmental sensors. The insulating plate may include an opening, and the blower may be positioned to draw air through the opening. The blower may include an inlet, and the blower may be positioned so that the inlet covers the opening.
- (6) One of the environmental sensors may be positioned to cover the opening.
- (7) The blower may be positioned on an opposite side of the bracket as the environmental sensors.
- (8) The two environmental sensors may each be positioned with a respective gap between the respective environmental sensor and the insulating plate, and the blower may be arranged to draw air through the gaps.
- (9) The sensor assembly may further include a cover extending across the environmental sensors, the cover and the insulating plate may define a chamber, the environmental sensors may be positioned in the chamber, and the blower may be positioned outside the chamber.
- (10) The insulating plate may be positioned between the environmental sensors and the bracket.
- (11) The insulating plate may abut the bracket.
- (12) The insulating plate may be shaped as a plate following an at least partially curved contour. The bracket may be shaped as a plate following the same contour as the insulating plate.
- (13) The sensor assembly may further include an air inlet positioned to direct air across a side of the insulating plate facing the environmental sensors.
- (14) The sensor assembly may further include a radome cover extending across the environmental sensors, the environmental sensors may be radar sensors, and the radome cover may be transparent to radio waves. The radome cover may be opaque to visible light.
- (15) The radome cover may be a vehicle fascia.
- (16) The radome cover may have a hydrophobic coating.
- (17) The bracket may be metal, and the insulating plate may be plastic.
- (18) The environmental sensors may be attached to the frame only via the bracket.
- (19) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a sensor assembly **102** includes a bracket **104** fixed to a frame **106** of a vehicle **100**, two environmental sensors **108** fixed to the bracket **104**, and an insulating plate **110** positioned between the environmental sensors **108** and an inside of the vehicle **100**.
- (20) The sensor assembly **102** may enhance the operation of the environmental sensors **108**. Mounting both environmental sensors **108** to the same bracket **104** can permit more stringent position tolerances to be met by the environmental sensors **108** compared to independently mounting the environmental sensors **108** to the frame **106** of the vehicle **100**. The tolerances for each environmental sensors **108** may include tolerances relative to the frame **106** as well as tolerances relative to the other of the environmental sensors **108**. Moreover, the bracket **104** can provide a reference for locating the insulating plate **110**. Additionally, the insulating plate **110** can isolate the environmental sensors **108** from heat generated by the vehicle **100**, e.g., the engine, permitting the environmental sensors **108** to more easily be kept within a temperature range for reliable operation. Finally, the same insulating plate **110** can be used for both environmental sensors **108**, in part due to the common position from the bracket **104**.

(21) With reference to FIG. 1, the vehicle **100** may be any passenger or commercial automobile such as a car, a truck, a sport utility vehicle, a crossover, a van, a minivan, a taxi, a bus, etc.

(22) The sensor assembly **102** includes a cover **112** extending across the environmental sensors **108**. The cover **112** can be transparent to a medium that the environmental sensors **108** are capable of detecting, and the cover **112** can form an exterior of the vehicle **100**. For example, if the environmental sensors **108** are radar sensors as described below, the cover **112** may be a radome cover that is transparent to radio waves. The cover **112** may be opaque to visible light to serve as an exterior of the vehicle **100**. The cover **112** may be a vehicle fascia, for example, may present a class-A surface, e.g., a finished surface exposed to view by a customer and free of unaesthetic blemishes and defects. As a vehicle fascia, the cover **112** may match a rest of a body of the vehicle **100**, thereby hiding the environmental sensors **108** from view. The cover **112** may include a hydrophobic coating on an exterior surface of the cover **112**, which may provide a clearer view for the environmental sensors **108**.

(23) The sensor assembly **102** may include an air inlet **114** positioned to direct airflow across the environmental sensors **108**. The air inlet **114** may be located on an exterior of the vehicle **100**. For example, the air inlet **114** may be an opening through the cover **112**. The air inlet **114** may be oriented in a vehicle-forward direction, e.g., on a front bumper as shown in FIG. 1, to permit ram air pressure through the air inlet **114**.

(24) The sensor assembly **102** may include a filter **116**. The filter **116** may extend across an air inlet **114**, e.g., so that air flowing through the air inlet **114** must pass through the filter **116**. The filter **116** removes solid particulates such as dust, pollen, mold, dust, and bacteria from the air flowing through the filter **116**. The filter **116** may be any suitable type of filter, e.g., paper, foam, cotton, stainless steel, oil bath, etc.

(25) With reference to FIG. 2, the vehicle **100** includes the frame **106**. The vehicle **100** may be of a unibody construction, in which the frame **106** and the body of the vehicle **100** are a single component. The vehicle **100** may, alternatively, be of a body-on-frame construction, in which the frame **106** supports the body that is a separate component from the frame **106**. The frame **106** and the body may be formed of any suitable material, for example, steel, aluminum, etc.

(26) The sensor assembly **102** may include the bracket **104** and a frame member **118** of the frame **106**. The bracket **104** is fixed to the frame **106**, e.g., the frame member **118**, of the vehicle **100**, for example, welded or fastened, e.g., bolted, riveted, etc.

(27) The bracket **104** is shaped as a plate following a contour, e.g., a same contour as the insulating plate **110** described below, which may facilitate compact packaging of the sensor assembly **102**. The contour is at least partially curved, which may facilitate packaging both of the environmental sensors **108** within a curved vehicle body. For example, the bracket **104** may include two bracket flat portions **120**, each bracket flat portion **120** having one of the environmental sensors **108** mounted thereto. The bracket flat portions **120** may be parallel to each other. The bracket **104** may also include a bracket step portion **122** connecting the bracket flat portions **120**, and the bracket step portion **122** may curve at the connection to each of the bracket flat portions **120**. The bracket flat portions **120** each have a length and a width, and the lengths and the widths are each greater than a distance between the bracket flat portions **120** perpendicular to the bracket flat portions **120**, which may facilitate packaging both of the environmental sensors **108** near an exterior of the vehicle **100**.

(28) The bracket **104** may be metal, e.g., steel. Using metal may help the bracket **104** be a highly durable and sufficiently rigid mounting location for the environmental sensors **108** so as to maintain the stringent tolerances for the environmental sensors **108**. When made of metal, the bracket **104** may have a high thermal conductivity, which may be counteracted by the insulating plate **110**, as described below.

(29) The insulating plate **110** may be attached to the bracket **104** and may abut the bracket **104**. For example, the insulating plate **110** may be fastened to the bracket **104** with fasteners **124** so that the

fasteners **124** are holding the insulating plate **110** against the bracket **104**. The same fasteners **124** may extend though and attach the bracket **104**, the insulating plate **110**, and the environmental sensors **108**. The fasteners **124** may be, e.g., bolts or rivets. By abutting the bracket **104**, the insulating plate **110** may provide compact packaging for the sensor assembly **102**.

(30) The insulating plate **110** is shaped to follow a contour, e.g., the same contour as the bracket **104**. The insulating plate **110** may be a lining of the bracket **104**. The contour is at least partially curved, which may facilitate packaging both of the environmental sensors **108** within a curved vehicle body. For example, the insulating may include two insulating flat portions **126**, each insulating flat portion **126** parallel to and abutting one of the bracket flat portions **120**. The insulating flat portions **126** may be parallel to each other. The insulating plate **110** may also include an insulating step portion **128** connecting the insulating flat portions **126**, and the insulating step portion **128** may curve at the connection to each of the insulating flat portions **126**. The insulating step portion **128** may contact the bracket step portion **122** from one of the insulating flat portions **126** to the other of the insulating flat portions **126**. The insulating flat portions **126** each have a length and a width, and the lengths and the widths are each greater than a distance between the insulating flat portions **126** perpendicular to the insulating flat portions **126**, which may facilitate packaging both of the environmental sensors **108** near an exterior of the vehicle **100**.

(31) The insulating plate **110** may be plastic. By being made of plastic, the insulating plate **110** may have a low thermal conductivity to protect the environmental sensors **108** from heat generated inside the vehicle **100**, e.g., by the engine. For example, the insulating plate **110** may have a thermal conductivity of at most 15 watts per meter-Kelvin (W/(m K)) at 25° C. When made of plastic, the insulating plate **110** may not provide structural support for the environmental sensors **108**, which may instead be provided by the bracket **104**, particularly if made of metal. Making the bracket **104** of metal and the insulating plate **110** of plastic thus provides synergistic benefits not provided by either material choice alone.

(32) With reference to FIG. 3, the cover **112** and the insulating plate **110** may define a chamber **130**. The insulating plate **110** may include edges **142** extending along and contacting the cover **112**. The edges **142** may prevent airflow between the cover **112** and the insulating plate **110**, i.e., airflow escaping the chamber **130**, at the edges **142**.

(33) The environmental sensors **108** detect the environment surrounding the vehicle **100**, e.g., objects and/or characteristics of the environment, such as other vehicles, physical barriers, pedestrians, etc. For example, the environmental sensors **108** may be radar sensors, ultrasonic sensors, scanning laser range finders, light detection and ranging (lidar) devices, or image processing sensors such as cameras. In particular, the environmental sensors **108** may be radar sensors, which are suited for being covered by the cover **112** that is opaque to visible light. As radar sensors, the environmental sensors **108** transmit radio waves and receive reflections of those radio waves to detect physical objects in the environment. The environmental sensors **108** can use direct propagation, i.e., measuring time delays between transmission and reception of radio waves, and/or indirect propagation, i.e., Frequency Modulated Continuous Wave (FMCW) method, i.e., measuring changes in frequency between transmitted and received radio waves.

(34) The environmental sensors **108** are positioned in the chamber **130**. The environmental sensors **108** are positioned directly between the cover **112** and the insulating plate **110**, e.g., outboard of the insulating plate **110** and inboard of the cover **112**. The environmental sensors **108** may be spaced from each other, e.g., vertically and/or longitudinally spaced from each other relative to a coordinate system of the vehicle **100**. Thus, each environmental sensor **108** may be positioned to have an unobstructed field of view extending through the cover **112**.

(35) The environmental sensors **108** are fixed to the bracket **104**, e.g., fastened to the bracket **104** with the fasteners **124**. The environmental sensors **108** may be attached to the frame **106** indirectly via the bracket **104**. For example, the environmental sensors **108** may be attached to the frame **106** only via the bracket **104**, i.e., the environmental sensors **108** are not independently attached to any

other component that is attached to the frame **106**. Being attached to the frame **106** only via the bracket **104** may help reliably fix the position of the environmental sensors **108** relative to the frame **106**.

(36) The insulating plate **110** and the bracket **104** are positioned between the environmental sensors **108** and the inside of the vehicle **100**, e.g., the engine of the vehicle **100**. The insulating plate **110** may be positioned between the environmental sensors **108** and the bracket **104**, thereby protecting the environmental sensors **108** from heat conducted through the bracket **104**. For example, the insulating plate **110** may be outboard of the bracket **104**, and the environmental sensors **108** may be outboard of the insulating plate **110**.

(37) Each environmental sensor **108** is positioned with a respective gap **132** between the respective environmental sensor **108** and the insulating plate **110**. The gaps **132** may provide further insulation for the environmental sensors **108**. Moreover, the thicknesses of the gaps **132**, i.e., the shortest distances between the insulating plate **110** and the environmental sensors **108**, may be sized to permit airflow between the insulating plate **110** and the environmental sensors **108**, thereby providing cooling for the environmental sensors **108**.

(38) The insulating plate **110** includes at least one opening **134**. The opening **134** may permit airflow to exit the chamber **130**, e.g., after entering the chamber **130** through the air inlet **114**. The opening **134** may be positioned partially or fully underneath one or both of the environmental sensors **108**, e.g., one of the environmental sensors **108** may be positioned to cover the opening **134**. This positioning may increase a quantity of airflow passing through the gap **132** underneath the environmental sensor **108**, increasing a cooling effect experienced by that environmental sensor **108**.

(39) Returning to FIG. 2, the sensor assembly **102** may include a blower **136**. The blower **136** increases the pressure of a gas by reducing a volume of the gas or by forcing additional gas into a constant volume. The blower **136** may be any suitable type of blower, e.g., a positive-displacement blower such as a reciprocating, ionic liquid piston, rotary screw, rotary vane, rolling piston, scroll, or diaphragm blower; a dynamic blower such as an air bubble, centrifugal, diagonal, mixed-flow, or axial-flow blower; or any other suitable type.

(40) The blower **136** may be fixed to the bracket **104**, thereby keeping a constant position relative to the opening **134**. For example, the blower **136** may be fastened to the bracket **104**. The blower **136** may be attached to the frame **106** indirectly via the bracket **104**. For example, the blower **136** may be attached to the frame **106** only via the bracket **104**, i.e., the blower **136** is not independently attached to any other component that is attached to the frame **106**. Having the blower **136** attached to the frame **106** only via the bracket **104** may facilitate installing most of the components of the sensor assembly **102** together as a unit to the frame member **118**. Alternatively to including the blower **136**, the sensor assembly **102** may rely on ram air pressure for drawing air through the air inlet **114** into the chamber **130**.

(41) The blower **136** includes a blower inlet **138** (shown in FIG. 3) and a blower outlet **140**. The blower **136** may be positioned so that the blower inlet **138** receives air from the chamber **130** and the blower outlet **140** exhausts air outside the chamber **130**, thereby removing heat absorbed by the air from the chamber **130**. For example, the blower **136** may be positioned outside the chamber **130** with the blower inlet **138** covering the opening **134**; thus, air exits the chamber **130** into the blower inlet **138**, travels through the blower **136**, and exits the blower **136** outside the chamber **130**.

Moreover, heat that transfers from the air to the blower **136** while passing through the blower **136** is already outside the chamber **130**, helping thermally isolate the environmental sensors **108**. For example, the blower **136** is positioned on an opposite side of the insulating plate **110** and the bracket **104** as the environmental sensors **108** are, e.g., inboard of the insulating plate **110** and the bracket **104**. The blower **136** can thereby be outside the chamber **130** while keeping the environmental sensors **108** with a view outside the vehicle **100** through the cover **112**.

(42) Returning to FIG. 3, the air inlet **114** is positioned to direct air into the chamber **130**, and the

blower **136** is positioned to draw air out of the chamber **130**. The airflow can provide cooling to the environmental sensors **108**. The air inlet **114** and the blower **136** can be positioned to draw air across the environmental sensors **108** to increase the rate of cooling. For example, the air inlet **114** may be positioned upstream from the environmental sensors **108**, and the blower **136** may be positioned downstream from the environmental sensors **108**. For example, the air inlet **114** may be positioned to direct air across a side of the insulating plate **110** facing the environmental sensors **108**, i.e., an outboard side, i.e., a side on an interior of the chamber **130**, thereby permitting airflow through the gaps **132** between the environmental sensors **108** and the insulating plate **110**. The blower **136** is arranged to draw air through the gaps **132** and through the opening **134**. The air heated by the environmental sensors **108** thus exits the chamber **130**, thereby removing heat from the chamber **130** and from the environmental sensors **108**.

(43) The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A sensor assembly comprising: a bracket fixed to a frame of a vehicle, the bracket shaped as a plate, the bracket having an exterior face and an interior face defined by the plate shape of the bracket; two environmental sensors mounted to the bracket on the exterior face; an insulating plate positioned between the environmental sensors and an inside of the vehicle; the insulating plate having a plate shape following the plate shape of the bracket; the insulating plate including an opening passing through the insulating plate; and a blower mounted to the bracket on the interior face, the blower including an inlet covering the opening along a direction perpendicular to the interior face at the opening; wherein one of the environmental sensors is positioned to cover the opening along a direction perpendicular to the exterior face at the opening.
2. The sensor assembly of claim 1, wherein the blower is positioned to draw air across the environmental sensors.
3. The sensor assembly of claim 2, wherein the blower is positioned on an opposite face of the insulating plate as the environmental sensors.
4. The sensor assembly of claim 3, wherein the blower is positioned to draw air through the opening.
5. The sensor assembly of claim 3, wherein the blower is positioned on an opposite face of the bracket as the environmental sensors.
6. The sensor assembly of claim 2, wherein the two environmental sensors are each positioned with a respective gap between the respective environmental sensor and the insulating plate, and the blower is arranged to draw air through the gaps.
7. The sensor assembly of claim 2, further comprising a cover extending across the environmental sensors, wherein the cover and the insulating plate define a chamber, the environmental sensors are positioned in the chamber, and the blower is positioned outside the chamber.
8. The sensor assembly of claim 1, wherein the insulating plate is positioned between the environmental sensors and the bracket.
9. The sensor assembly of claim 1, wherein the insulating plate abuts the bracket.
10. The sensor assembly of claim 1, wherein the plate shape of the insulating plate follows an at least partially curved contour.
11. The sensor assembly of claim 10, wherein the plate shape of the bracket follows the same contour as the insulating plate.
12. The sensor assembly of claim 1, further comprising an air inlet positioned to direct air across a face of the insulating plate facing the environmental sensors.

13. The sensor assembly of claim 1, further comprising a radome cover extending across the environmental sensors, wherein the environmental sensors are radar sensors, and the radome cover is transparent to radio waves.
 14. The sensor assembly of claim 13, wherein the radome cover is opaque to visible light.
 15. The sensor assembly of claim 13, wherein the radome cover is a vehicle fascia.
 16. The sensor assembly of claim 13, wherein the radome cover has a hydrophobic coating.
 17. The sensor assembly of claim 1, wherein the bracket is metal, and the insulating plate is plastic.
 18. The sensor assembly of claim 1, wherein the environmental sensors are attached to the frame only via the bracket.
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